BASKET · DESIGN SPECIFICATION

Motion & Interaction Spec

Every state, transition, gesture and haptic in the app — with Flutter durations, curves and widget notes. Professional but playful: motion is quick and confident, with just enough overshoot to feel alive.

PERSONALITY	HAPTICS	PLATFORM	COMPANION FILES
Springy & playful	**Generous**	**Flutter · iOS + Android**	**app_colors.dart · app_theme.dart**

01 — PRINCIPLES

Five rules that make it feel alive

1 · Motion confirms, never decorates

Every animation answers "did that work?" or "where did it go?". If it answers neither, cut it. No looping ambient motion, no decorative parallax.

2 · Overshoot is the personality

Things the user *did* get a small overshoot (checks, chips, added rows). Things that merely *happen* (loads, syncs, someone else's edit) stay linear and calm.

3 · Fast in, gentle out

Appearing is 180–260 ms. Disappearing is 120–160 ms. Nothing in the app takes longer than 400 ms except the completion celebration.

4 · Colour follows the shop

Accent motion (checks, tab pills, FAB) tints to the active shop's user-chosen colour via `AppTheme.forShop`. Structure stays ink.

5 · One spring, one curve, one duration set

Everything in this document is built from the six tokens on the next page. If a screen needs a seventh, it is probably wrong.

02 — TOKENS

Duration & curve tokens

TOKEN	DURATION	CURVE	USED FOR
`micro`	120 ms	`Curves.easeOut`	Press states, ripples, opacity flips, icon swaps
`quick`	180 ms	`Curves.easeOutCubic`	Chip selection, colour tint, row strike-through, dismissals
`standard`	260 ms	`Curves.easeOutCubic`	Page/tab transitions, drawer, list reflow
`bouncy`	340 ms	`Curves.easeOutBack`	Checkbox fill, newly added row, FAB, badge pops
`sheet`	380 ms	`Curves.easeOutQuint`	Bottom sheets in; 220 ms `easeInCubic` out
`celebrate`	900 ms	`SpringDescription(mass: 1, stiffness: 180, damping: 14)`	List-complete moment only

app_motion.dart

```
import 'package:flutter/material.dart';
import 'package:flutter/physics.dart';

class AppMotion {
  AppMotion._();

  static const Duration micro     = Duration(milliseconds: 120);
  static const Duration quick     = Duration(milliseconds: 180);
  static const Duration standard  = Duration(milliseconds: 260);
  static const Duration bouncy    = Duration(milliseconds: 340);
  static const Duration sc-camel-sheet-in
      = Duration(milliseconds: 380);
  static const Duration sc-camel-sheet-out
      = Duration(milliseconds: 220);
  static const Duration celebrate = Duration(milliseconds: 900);

  static const Curve enter   = Curves.easeOutCubic;
  static const Curve exit    = Curves.easeInCubic;
  static const Curve pop     = Curves.easeOutBack;   // the "playful" one
  static const Curve sheet   = Curves.easeOutQuint;
  static const Curve press   = Curves.easeOut;

  static const SpringDescription spring =
      SpringDescription(mass: 1, stiffness: 180, damping: 14);

  /// Stagger delay for list children entering together (cap at 6 items).
  static Duration stagger(int index) =>
      Duration(milliseconds: (index.clamp(0, 5)) * 35);
}
```

Reduced motion. Wrap every non-essential animation in a check on `MediaQuery.of(context).disableAnimations`. When true: durations collapse to `micro`, `easeOutBack` falls back to `easeOut`, and the completion celebration becomes a static banner.

03 — HAPTICS

Haptic map

Generous, but never twice for one action. Rule: haptics fire on *commitment*, not on touch-down.
Import `package:flutter/services.dart`.

MOMENT	CALL	NOTE
Check an item off	`HapticFeedback.lightImpact()`	On check only, not on uncheck
Shop tab settles after swipe	`HapticFeedback.selectionClick()`	Fire on index change, not per frame
Product added	`HapticFeedback.lightImpact()`	As the row lands, not on submit tap
Long-press enters multi-select	`HapticFeedback.mediumImpact()`	The clearest mode-change signal in the app
Drag pick-up / drop	`selectionClick()` / `lightImpact()`	Also on each reorder-hover crossing
Swipe passes delete threshold	`HapticFeedback.mediumImpact()`	At threshold crossing – pre-commit warning
List completed (last item checked)	`HapticFeedback.heavyImpact()`	Replaces the light impact for that tap
Error (bad code, failed save)	`HapticFeedback.vibrate()`	Paired with the shake animation
Onboarding step advances	`HapticFeedback.selectionClick()`	Light punctuation per step

04 — SIGNATURE MOMENT

Checking an item off

The single most repeated action in the app. It gets the most craft. Four things animate at once, all driven from one 340 ms controller.

LAYER	FROM → TO	TIMING
Circle fill	scale 0 → 1 of the shop-coloured disc, radial from centre	340 ms `easeOutBack`
Checkmark path	stroke draw 0 → 1 (dash offset)	200 ms `easeOutCubic`, 80 ms delay
Row content	opacity 1 → .45; strike-through wipes left→right	180 ms `easeOutCubic`
Row itself	scale 1 → .985 → 1 (a subtle "press down")	240 ms, symmetric

Then, 600 ms later: the row animates out of its position and re-inserts at the bottom of the list under a "Done · 4" divider – `standard` 260 ms, driven by `AnimatedList` or `flutter_animate`. The delay matters: reordering instantly makes people feel they mis-tapped. Unchecking reverses everything at `quick` with no overshoot and no reorder delay.

Strike-through wipe

```
// Don't toggle TextDecoration.lineThrough — it snaps. Draw it.
Stack(children: [
  Text(item.name, style: nameStyle),
  Positioned(
    left: 0, right: 0, top: 0, bottom: 0,
    child: AnimatedBuilder(
      animation: _strike,               // 0 → 1 over AppMotion.quick
      builder: (_, __) => CustomPaint(
        painter: StrikePainter(progress: _strike.value, color: mutedColor),
      ),
    ),
  ),
]);
```

05 — SIGNATURE MOMENT

Swiping between shop tabs

The tab strip and the page view are driven by one `TabController`, so the pills react continuously to the drag rather than snapping at the end. This continuous coupling is what separates it from a stock tab bar.

ELEMENT	BEHAVIOUR DURING DRAG
Active pill background	Colour lerps between the two neighbouring shop colours by `_tabController.animation!.value` fraction — mid-swipe you see the blend
Pill scale	Outgoing pill 1 → .96, incoming .96 → 1, same fraction
Pill label weight	Cross-fade two `Text` layers (w400/w600) rather than animating weight
Strip scroll	Auto-centres the active pill, 260 ms `easeOutCubic`
Quick-add bar dot	Tints to the incoming shop colour on the same fraction
Settle	`selectionClick()` haptic the moment `indexIsChanging` resolves

```
// Rebuild pills continuously off the controller's animation.
AnimatedBuilder(
  animation: _tabController.animation!,
  builder: (context, _) {
    final t = _tabController.animation!.value;    // e.g. 1.62 mid-swipe
    return Row(children: [
      for (var i = 0; i < tabs.length; i++)
        ShopTabPill(
          shop: tabs[i],
          selectedness: (1 - (t - i).abs()).clamp(0.0, 1.0), // 0 → 1
        ),
    ]);
  },
);
```

The trailing + button sits outside the `TabBar` in the same `Row`, so it never enters the swipe sequence. Bounds: swiping past the last shop or before "All" gets the platform overscroll glow — no wrap-around.

06 — SIGNATURE MOMENT

Adding a product

The whole point is continuity: the sheet doesn't just close, it *becomes* the row. Sequence, from the tap on "Add":

T	WHAT HAPPENS	SPEC
0 ms	Sheet slides down and out	220 ms `easeInCubic`, scrim fades over 160 ms
80 ms	List makes room — following rows slide down by the new row's height	260 ms `easeOutCubic`, via `AnimatedList.insertItem`
140 ms	New row fades + scales in at its slot	opacity 0→1, scale .94→1, 340 ms `easeOutBack`
140 ms	Shop dot pops	scale 0→1, 340 ms `easeOutBack`, 60 ms extra delay
300 ms	Row background flashes the shop colour at 8% and fades out	Hold 200 ms, fade 400 ms `easeOut`
340 ms	`lightImpact()`	As the row lands, so it feels like it hit the list

Adding from the "All" tab needs a shop first: tapping the quick-add bar there expands it to show a one-row shop chooser above the field, 260 ms `easeOutCubic` height animation. **Rapid entry:** the sheet stays open if the user submits with the keyboard's "next" action — field clears, row lands behind the sheet, so a whole list can be typed in one sitting.

07 — SIGNATURE MOMENT

Onboarding step transitions

A `PageView` with `NeverScrollableScrollPhysics` — button-driven only, so the transition is fully authored.

ELEMENT	FORWARD TRANSITION
Outgoing page	slide −24 px + fade to 0 over 200 ms `easeInCubic`
Incoming page	slide +32 px → 0 + fade in, 320 ms `easeOutCubic`, 60 ms delay
Incoming content	title, body, field stagger 35 ms apart via `AppMotion.stagger`
Step counter	"Step 1 of 3" number slides up and cross-fades; a 3-dot rail below fills the active dot 6 px→20 px pill, 340 ms `easeOutBack`
Continue button	Stays fixed — never re-animates. Label cross-fades to "Done" on the last step, 180 ms
Back	Mirror image, and the step-rail pill shrinks rather than jumping

Validation shake. Continue with an empty required field: field translates ±6 px, 4 oscillations over 320 ms `easeOut`, border lerps to `AppColors.destructive` over 120 ms, error text slides down 8 px + fades in, `HapticFeedback.vibrate()`. Error clears on the next keystroke — border and text fade out over 180 ms, no shake.

Welcome → onboarding. A route push, but the Basket mark is a `Hero` that shrinks and moves to the top-left corner as the first step arrives, 380 ms `easeOutQuint`.

08 — SIGNATURE MOMENT

Live edits from other people

The reason the app is shared — so it should be visible, but never startling. Remote changes are *calm*: no overshoot, no haptic, no sound.

EVENT	TREATMENT
Someone adds an item	Row slides in from its slot with a 1.5 px left edge in that person's avatar colour; a small avatar chip fades in on the right for 3 s, then fades to the qty text. 300 ms `easeOutCubic`, no scale pop.
Someone checks an item	Same check animation but at 180 ms with `easeOutCubic` (no back-curve), and no haptic. Avatar chip appears beside the check for 3 s.
Someone removes an item	Row fades to 30%, holds 400 ms, then collapses its height over 220 ms — the pause prevents "the list jumped".
Person joins the list	Their avatar drops into the header cluster, 340 ms `easeOutBack`, plus a 2.5 s toast: "Sam joined Weekly Groceries".
Presence	Avatars of people currently viewing sit in the header at full opacity; recent-but-away at 40%. Transition 400 ms linear.
Conflict	Last write wins silently, except for quantity: show an inline "Sam set this to 3" caption under the row for 4 s.

Never animate a remote change while the user is dragging, multi-selecting, or has a sheet open — queue it and apply on the next idle frame, so their gesture is never yanked out from under them.

09 — SIGNATURE MOMENT

Completing the whole list

The one place the app is allowed to be theatrical — and only when the *last unchecked item in the whole list* is checked, not per shop tab.

T	BEAT
0 ms	Normal check animation runs, but the haptic is upgraded to `heavyImpact()`
240 ms	All checked rows' shop dots pulse outward in sequence, 40 ms apart, scale 1 → 1.35 → 1
400 ms	A card rises from the bottom over the quick-add bar: "All done — 14 items". Spring in (`AppMotion.spring`), so it settles with one soft bounce
400 ms	Confetti, but restrained: 14–20 particles in the list's shop colours only, gravity fall, 1.4 s, no repeat. Skip entirely if reduced-motion is on
Idle	Card offers "Clear checked items" and "Keep list" — it does *not* auto-dismiss; tapping anywhere else slides it away over 220 ms

Fires at most once per list-completion event; re-checking an item after clearing does not re-fire until the list has been incomplete again.

10 — GESTURES

Swipe, long-press, drag, pull

Swipe row to delete

`Dismissible`, `direction: endToStart` only (left-to-right stays free for a future "move to shop"). Background is `AppColors.destructive` with a trash icon that scales 0.7 → 1 as the drag passes 30% width. `dismissThresholds: {endToStart: 0.4}`. At threshold crossing: `mediumImpact()` and the icon pops to 1.15 then settles. On release below threshold, the row springs back with `AppMotion.spring`. On dismiss, height collapses over 220 ms and a 4-second undo snackbar appears — the row is only deleted from Firebase when the snackbar expires, so undo is instant and local.

Long-press to multi-select

500 ms press. At 300 ms the row starts scaling to 0.97 as a "keep holding" tell; at fire, `mediumImpact()` and the row springs back to 1.0. Entering the mode: every row's checkbox slides 8 px right and a selection circle fades in on the left (240 ms, 25 ms stagger down the list), the header cross-fades to a contextual bar ("3 selected · Move · Delete · ✕"), and the quick-add bar slides out. Exiting reverses it. Tapping rows toggles selection with `selectionClick()` and a 120 ms fill.

Drag to reorder

`ReorderableListView` within a single shop tab (cross-shop reordering is meaningless — moving between shops is a swipe action instead). Pick-up: row lifts to `elevation 8`, scales to 1.03, opacity 0.95, over 200 ms, with `selectionClick()`. Other rows part with a 200 ms `easeOutCubic` gap. Each crossing fires `selectionClick()`. Drop: settles with `AppMotion.spring` and `lightImpact()`. Checked items are not draggable and sit locked below the "Done" divider.

Pull to refresh

`RefreshIndicator` tinted to the active shop colour (`colorScheme.primary`). Since Firestore streams are already live, this is a *reassurance* gesture: it forces a re-fetch and, on completion, shows a 1.2 s "Up to date · just now" caption that fades in under the tab strip. Minimum spinner time 600 ms even if the fetch returns instantly — an instantly-vanishing spinner reads as broken.

11 — MISSING FLOW

Empty states

Four distinct empties. Each is a centred stack: a light line-illustration slot (48 px, 12% ink), a title in `bodyStrong`, one line of `secondary` copy, and at most one action. All fade+rise in (12 px, 300 ms `easeOutCubic`) so switching to an empty tab isn't a blank flash.

WHERE	TITLE	BODY & ACTION
Brand-new list	"Nothing on the list yet"	"Add your first product below." An animated arrow nudges toward the quick-add bar — 3 gentle bounces, then stops for good.
Empty shop tab	"Nothing for Colruyt"	"Items you add here will show on this tab." Action: *Add a product*. Uses the shop's colour for the illustration tint.
All items checked	"All done"	Not an empty state — see §09. Rows stay visible under the Done divider with a *Clear checked* action.
No lists at all	"No lists yet"	Only reachable after deleting the last list. Actions: *New list* and *Join a list*.
No shops on a list	"Add a shop to get started"	The tab strip collapses to just the +, which pulses once every 4 s until tapped.
Search, no results	"No products match ‘milk’"	Action: *Add "milk" to this list* — turns a dead end into the fast path.

12 — MISSING FLOW

Checked items & the Done section

Checked items never disappear on their own — the shopper needs to see what's in the trolley.

RULE	BEHAVIOUR
Position	Checked rows move below a sticky `Done · 4` divider, 600 ms after being checked
Appearance	45% opacity, strike-through, shop dot desaturated to 50%
Collapse	The Done divider is tappable — collapses the section with a 260 ms height animation and a chevron that rotates 180°. Collapsed state persists per list.
Clear	*Clear checked* in the divider. Rows fly out staggered 40 ms apart (fade + 16 px right, 200 ms each), then the section collapses. Undo snackbar, 5 s.
Uncheck	Row animates back up into its shop group immediately — no 600 ms delay, since intent is unambiguous
Counts	Each tab pill shows a remaining count that ticks down with a 120 ms slide-up digit change; it hides at zero rather than showing "0"

13 — MISSING FLOW

Editing & deleting lists

ACTION	ENTRY POINT	BEHAVIOUR
Rename	Tap the list title in the header, or long-press in the drawer	Title becomes an inline field in place (no sheet) — cursor at end, 180 ms border fade-in. Saves on submit or blur; the header title cross-fades to the new name.
Reorder lists	Drawer, long-press a list	Same drag spec as §10, scoped to the drawer
Archive	List overflow menu	Hides from the drawer, keeps data. A footer link "2 archived" restores.
Delete (owner)	List overflow menu	Confirm dialog naming the list and its member count: "Delete Weekly Groceries? This removes it for all 3 members." Destructive button. No undo — this one is genuinely permanent, so it earns a dialog rather than a snackbar.
Leave (member)	List overflow menu	Non-owners see *Leave list* instead of Delete. Confirm sheet, then route to the next list with a 260 ms cross-fade.
After deletion	–	Drawer entry collapses (220 ms), app switches to the next list; if none remain, the "No lists yet" empty state.

14 — MISSING FLOW

Editing & deleting shops

Shops are user-created, so their colours and names must be fully editable — and deleting one must never silently destroy products.

ACTION	BEHAVIOUR
Edit shop	Tap a row in Manage shops → the same sheet as Add a shop, pre-filled, button reads *Save changes*
Change colour	Selecting a swatch retints *live behind the sheet*: the tab pill, every dot for that shop, and the accent theme lerp over 300 ms. This preview is the best part of the flow — the user sees the consequence before committing.
Colour clash	If the chosen colour is within a small perceptual distance of an existing shop's, show an inline caption "Close to Colruyt" — a warning, never a block
Reorder shops	Drag handle in Manage shops; the tab strip re-orders live behind the screen
Delete a shop with items	Sheet: "Colruyt has 6 products." Two options — *Move them to…* (shop picker) or *Delete products too* (destructive). Never a bare confirm.
Delete an empty shop	Immediate, with a 5 s undo snackbar. The tab pill collapses horizontally over 220 ms and the strip closes the gap.
Scope	Shops belong to the household, not one list — the sheet states "Used on 2 lists" so deletion is never a surprise elsewhere

15 — MISSING FLOW

Household members & permissions

Deliberately shallow — two roles, no admin console. A shopping list does not need an org chart.

ROLE	CAN
Owner (creator)	Everything, plus: rename/delete the list, remove members, regenerate the invite code, transfer ownership
Member	Add/edit/check/delete products, add and edit shops, invite others, leave the list

Members screen (Settings → Invite & sharing): avatar + name rows, "Owner" tag on the first, "You" tag on yourself, and a last-active caption. Row overflow (owner only): *Make owner*, *Remove from list*. Removal shows a confirm sheet and the row collapses over 220 ms.

Invite. The code card has a *Copy* button that morphs to a checkmark for 1.6 s (icon cross-fade 180 ms, `lightImpact()`), plus *Share* (native sheet) and a QR toggle that flips the card over on the Y-axis, 420 ms `easeOutQuint`. Codes expire after 7 days; an expired code shows the state below, with *Generate new code*.

Joining. Successful code entry: field border flashes green, the sheet content cross-fades to a 400 ms "Joined Weekly Groceries" confirmation with the list's shop dots animating in, then routes to the list. The new member's avatar animates into every other member's header live.

16 — MISSING FLOW

Search & suggestions

Search lives behind an icon in the header, not a permanent bar — the list is usually short. Tapping it expands the field across the header (280 ms `easeOutCubic`, title fades out), the tab strip slides up and out, and results become one flat list across all shops with the shop dot doing the disambiguating. Matching substrings are weight-600 rather than highlighted. Clearing collapses everything back. Search is local-only — no spinner, results filter per keystroke.

Suggestions appear inside the add-product sheet, under the field: a wrap of chips from (1) items bought on this list in the last 30 days, (2) items typed most often, ranked by recency × frequency. Chips stagger in 30 ms apart. Tapping one adds it immediately and the chip flies to the list, leaving the sheet open for the next. With text typed, the chip row becomes matching-history autocomplete instead. Suggestions are per household, not global — no shared product catalogue, nothing to moderate.

Recently bought is the same data on a dedicated screen (drawer → Recently bought): grouped by week, each row with a `+` that re-adds to the current list with a `easeOutBack` pop and `lightImpact()`. This is the "weekly staples" shortcut without building a template feature.

17 — MISSING FLOW

Loading & skeletons

Never a centred spinner on a screen that has known structure. Rule of thumb by expected wait:

WAIT	TREATMENT
< 300 ms	Show nothing. A flashed skeleton is worse than a beat of stillness.
300 ms – 2 s	Skeleton rows: 4 placeholder rows matching the real row geometry (22 px circle, 7 px dot, 60% width bar), ink at 6%, shimmer sweeping left→right over 1.4 s, `easeInOut`, infinite. Real rows cross-fade in over 220 ms with a 30 ms stagger.
> 2 s	Keep the skeleton and add a caption: "Still loading…" then, at 6 s, "Check your connection" with a *Retry*.
In-button	Primary buttons (Join, Create) collapse their label and show an 18 px indicator in `onPrimary`; the button keeps its width so nothing reflows. Minimum 400 ms visible.
Optimistic writes	Adding, checking and renaming apply instantly at 100% opacity with no pending style. If the write fails, revert with a 200 ms fade and a "Couldn't save · Retry" snackbar. Optimism beats spinners for a list app.

18 — MISSING FLOW

Offline & sync

Supermarkets have terrible signal, so offline is a primary state, not an error. Firestore's local cache does the work; the UI's job is honesty without anxiety.

STATE	TREATMENT
Goes offline	A 28 px bar slides down under the header (240 ms `easeOutCubic`), subtle ink-8% background: "Offline — changes will sync". No colour alarm, no icon jitter.
Editing offline	Fully allowed. Rows written offline carry a 1.5 px dashed left edge instead of solid — a quiet "not yet confirmed" mark.
Reconnects	Bar text cross-fades to "Syncing…" with a 2 px indeterminate progress line, then to "Up to date" in green-ink for 1.6 s, then slides away. Dashed edges resolve to solid, staggered 40 ms apart.
Sync conflict	Silent last-write-wins, except deletions of items someone else edited — those surface a single "1 change couldn't be applied" snackbar with *Review*.
Long offline (> 24 h)	On reopen, show the list from cache immediately with the offline bar reading "Showing your last saved copy"

19 — MISSING FLOW

Errors

Three tiers, chosen by whether the user can act: **inline** (their input is wrong), **snackbar** (something failed but the app is fine), **screen** (nothing can be shown).

CASE	TIER	MESSAGE & MOTION
Empty required field	Inline	Shake + red border + helper text (§07)
Invalid invite code	Inline	"That code doesn't match any list." Shake, `vibrate()`, field keeps the text so it can be corrected rather than retyped.
Expired invite code	Inline	"This invite has expired — ask for a new one." No shake; it isn't a typo.
Already a member	Inline	"You're already on this list." Action: *Open it*.
Duplicate shop name	Inline	"You already have a shop called Lidl."
Write failed	Snackbar	"Couldn't save · Retry". Optimistic row reverts with a 200 ms fade.
Permission denied	Snackbar	"Only the owner can do that." Fires when a stale UI allows a now-forbidden action.
List deleted while open	Screen	Full-screen: "This list was deleted." One action, *Back to your lists*. Cross-fade 300 ms — never an abrupt pop.
Removed from list	Screen	"You no longer have access to this list."
Unexpected crash	Screen	"Something went wrong." *Try again* + *Report*. Never a raw exception string.

All snackbars: floating, 12 px radius, 16 px margins, slide up 240 ms `easeOutCubic`, 4 s (5 s with an undo), swipe-to-dismiss, and they lift above the quick-add bar rather than covering it.

20 — BUILD ORDER

What to build first

Roughly 80% of the perceived liveliness comes from the first four rows. Do them before anything below the line.

#	ITEM	SECTION	EFFORT
1	Check-off animation + haptic	§04	Half a day — highest return in the app
2	Continuous tab-pill coupling	§05	Half a day, one AnimatedBuilder
3	Add-product insertion sequence	§06	A day with AnimatedList
4	Empty states + skeletons	§11, §17	A day; removes all the "broken?" moments
5	Swipe-delete with undo	§10	Half a day, Dismissible
6	Offline bar + sync states	§18	A day, mostly plumbing
7	Onboarding transitions + shake	§07	Half a day
8	Done section + clear checked	§12	A day
9	Edit/delete lists & shops	§13, §14	Two days including the move-products case
10	Live-edit presence & avatars	§08	Two days
11	Search + suggestions	§16	Two days
12	Multi-select + reorder	§10	Two days — the most optional items here
13	Completion celebration	§09	Half a day, best saved for last